

Lab 1: Requirements Engineering & UML Use-Case

Modelling

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall validate proposed pallet placement coordinates against the rack maximum load capacity before confirming storage.	High	Pass: Pallet is assigned to a valid rack without exceeding its maximum load capacity. Fail: Placement is permitted when the total weight exceeds the rack's rated structural limit.	Prevents storage on overloaded racks and ensures safe pallet placement.
FR-002	Functional	The system shall allow the warehouse operator to assign a pallet to a specific 3D shelf position.	High	Pass: A pallet is assigned to a valid 3D shelf position and its coordinates are recorded. Fail: The system accepts an invalid or undefined shelf position.	Ensures accurate tracking of pallet locations within the warehouse.
FR-003	Functional	The system shall allow authorized users to retrieve the stored location of a pallet using its SKU or pallet identifier.	High	Pass: Entering a valid SKU or pallet ID displays the correct stored shelf coordinates. Fail: The system displays an incorrect location or no	Enables users to quickly locate pallets in the warehouse.

				location for a valid pallet.	
FR-004	Functional	The system shall record pallet stock movements when a barcode or RFID scan is successfully received.	High	Pass: A valid barcode/RFID scan creates a stock movement record with the pallet identifier and updated location. Fail: A valid scan occurs but no movement record is created.	Maintains an accurate record of pallet movements.
FR-005	Functional	The system shall allow the logistics supervisor to view pallet locations and stock movement records.	Medium	Pass: The logistics supervisor can view current pallet locations and stock movement records. Fail: The supervisor cannot access the required tracking information.	Helps the supervisor monitor warehouse inventory and movements.
NFR-001	Nonfunctional	The warehouse item lookup service must locate any SKU pallet coordinate across 100,000 bins in under 100 ms.	High	Pass: The system locates the requested SKU pallet coordinate within 100 ms across 100,000 bins under simulated peak load. Fail: The lookup takes 100 ms or longer.	Ensures fast inventory lookup in a large warehouse.
NFR-002	Nonfunctional	The system shall	High	Pass: Authorized	Protects warehouse

		restrict inventory and stock movement records to authorized warehouse users based on their assigned role.		users can access permitted inventory records, while unauthorized users are denied access. Fail: An unauthorized user can access restricted records.	inventory and movement information from unauthorized access.
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